

Theodor Vararu

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About

I'm Theo, a software engineer with a passion for accessibility and progressive enhancement.

I specialise in shipping full-stack web applications, and love to dabble in design, local ML, AI-assisted development, Linux, and entrepreneurship. I'm currently working on a national vaccination platform for the UK government, and building my own ideas on the side.

I like computers, metal, philosophy, minimalism, and road trips. I maintain a menagerie of pet servers, including my home lab and SFF rig. I have a Bachelor's degree in Engineering.

Experience

National Health Service (Contract), Tech Lead, Jan 2023 to Apr 2026

Led development of the **Manage vaccinations in schools** (Mavis) service, a national service used by nurses to record consent and vaccinations. I took the project from initial commit to millions of patients.

Department for Education (Contract), Tech Lead, Jan 2022 to Dec 2022

Led development of 4 projects across the **Teaching Regulation Authority** platform, including a OIDC SSO provider, and a service to manage serious misconduct.

Ministry of Justice (Contract), Senior Developer, Sep 2021 to Dec 2021

Brought in to facilitate an **accessibility audit** of the **Manage Supervisions** service within the HMPPS platform. Carried out further improvements and fixed accessibility issues.

Meta, Software Engineer, Jan 2021 to Jul 2021

Worked on the **Workplace** product as part of the **Web Core** team, in the **Accessibility** squad. Triaged, identified, and fixed accessibility issues in the push towards full **WCAG 2.2** compliance.

Helped the team refine their strategy and roadmap, by writing an **accessibility statement**, gathering information from previous audits, and organising existing tasks and work streams.

Department for Education (Contract), Tech Lead, Sep 2019 to Nov 2020

Led development of the candidate-facing side of Apply for teacher training in **Ruby on Rails**, **JavaScript**, and **PostgreSQL**, on **Microsoft Azure**. Collaborated with stakeholders, architects, and other tech leads towards an MVP public beta release in November 2019, which was achieved.

Worked in an **agile** team, **unblocking**, **pairing** with, and **mentoring** 3 to 5 developers of varying seniority levels. Drove conversations with the **Find** team to **collaborate** on shared features.

Encouraged team participation in the **design** and **user research** process, improved the pull-request **process** by introducing review apps, and handled support requests and production incidents.

Championed **accessibility**, frontend **performance**, and **automated testing** by prioritising related improvements into our **sprints**.

Department for Education (Contract), Senior Developer, Sep 2018 to Sep 2019

Developed full-stack features as part of the Becoming a Teacher team, in **C#**, **Ruby on Rails**, **PostgreSQL**, and **JavaScript / Node.js**, to help teachers Find postgraduate training courses.

Improved the **DevOps** pipeline by adopting **Docker**, reducing build times, using **Terraform**, and adding **end-to-end smoke tests** using Cypress.io, on both Travis CI and **Microsoft Azure**.

Developed prototypes using **Next.js**, **Now.sh**, and **Apollo.js** to trial **Node.js**, **Serverless**, and **GraphQL** for our services. Findings were presented at the periodical development and architecture meeting, to favourable reception from technical leaders.

Took part in **Agile** ceremonies, wrote specifications for full-stack features, and collaborated / paired on a number of different problems across the stack.

Gamesys (Contract), Senior Developer, Mar 2018 to Sep 2018

Developed user-facing features on the **React.js**, **Node.js**, and **Flux**-based single-page application frontend, focused on addressing user experience issues, industry-specific regulatory requirements, and troubleshooting Selenium-based tests.

Built an internal tool for managing web chat emoji packs, using **Next.js** (static mode with server-side rendering), Cypress, and Tachyons.io. Leveraged client-side JavaScript to cut/stitch PNG assets and package into .zips without needing a server. MVP delivered in record time to a high standard.

Worked on optimising **frontend performance**. Removed the project's dependency on **moment.js**, and trimmed down the dependency on **lodash**, resulting in total savings of **33%** on the bundle size. Wrote performance **documentation** to aid with future work.

Provided **mentoring** to team members and an apprentice, focusing on a pragmatic but **users-first** approach to software engineering. Facilitated agile ceremonies and interviews for frontend Node.js as well as backend Ruby candidates.

Graphit Systems (Contract), Senior Developer, Aug 2017 to Feb 2018

Built user-facing features for the online 3D configurator. Shipped a redesign, online payment with **Stripe**, fixes for **legacy browsers** and **performance** issues while building on top of the previous developer's work without handover. **React.js**, **Three.js**, **webpack**, **Redux.js**.

Collaborated with the designer to create a revised **styleguide**, consolidating rhythm and colours throughout the app, backed by a Tachyons-inspired atomic CSS **design system**.

Championed **user research**-based decision-making, highlighted and fixed **accessibility** pitfalls, and provided input into design discussions to reach solutions that are **mobile-first** and **performant**.

Improved product quality by implementing **end-to-end testing** using Cypress, Jest.js component tests, Sentry.js error tracking, and a bespoke gettext and Babel.js **internationalisation** pipeline.

Delivered a proof-of-concept React Native and Expo.io native AR application in only 3 days by reusing and adapting existing 3D code from the main codebase.

Government Digital Service, Senior Developer, Mar 2016 to Aug 2017

Led development of a reusable country picker component. Talked to service teams to learn about their needs, prototyped initial implementation, built an Accessible Autocomplete component and a suggestion engine. Took the component to user research and to an accessibility audit from the Digital Accessibility Centre. **Preact.js**, **TDD**, **Selenium testing**, **accessibility testing**, **progressive enhancement**, **webpack**.

Built a small **JSON API** to generate country picker data based on OpenRegister data. **Java**, **Serverless**, **AWS**, **TDD**, **JSON**.

Sole developer on the Webchat Alpha project, building user research prototypes and testing with users. Focused on **accessibility** of webchat interfaces. Released findings as presentations and blog posts. **Node.js**, **React.js**, **Socket.io**, **accessibility testing**.

Led development of a **data visualisation** prototype to view digital transformation dependencies. **Node.js**, **React.js**, **D3.js**, **Google Sheets API**.

Rebuilt GOV.UK formats as part of the Migration programme team, moving from the legacy monolith publisher to a **microservices** pipeline. **Ruby on Rails**, **TDD**, **Agile**.

Presented talks about **React.js** and **accessibility**. Provided mentoring to apprentices, line-managed 2 developers.

Timecounts, Senior Developer, May 2015 to Feb 2016

Built product features, collaborating with stakeholders to define specifications. **Node.js**, **React.js**, **TDD**, **Ruby on Rails JSON API**, **SASS / CSS modules**.

Major features shipped include a calendar view and date picker component, CSV import validation, landing and marketing pages, **animations**, improvements to the build process, and backend support fixes.

Helped the team refine their **Agile** processes, implemented code review and issue tracking methods. Championed **functional programming** and linting.

Workangel, Lead Developer, Sep 2014 to Apr 2015

Built product features on the admin panel and user-facing application, two single-page applications that interact with a **RESTful JSON API**. **Node.js**, **Angular.js 1.x**, **Stylus**.

Shipped a **mobile-first design** rebuild and **animations** for inter-page transitions, both unscoped but delivered to extremely favourable reception.

Dramatically improved the build process by **optimising the performance** of tasks and embracing modern best practices: template minification, linting, unit and Selenium integration testing, continuous integration, load testing.

Defined Jade, Stylus, and JavaScript styleguides. Provided mentoring and coaching to new developers.

ReadForward, Lead Developer, Mar 2013 to Sep 2014

Led development of an open-source digital textbook authoring tool. Built the pipeline and the frontend pattern library. Onboarded and trained junior developers in its use that produced our digital textbook offerings for primary education. **Node.js**, **Backbone.js**, **pattern libraries**, **visual regression testing**.

Architected and managed a tour guide app for the Danube Delta. Delivered 1.0 in just **3 weeks** of development time. **Ampersand.js**, **Famous.js**, **Cordova**, **testing**, **mapping and route generation**, **offline-first**.

Architected and led initial development of the open-source website for Monica Macovei's 2014 presidential campaign. Scored 100/100 on Google PageSpeed Insights using **performance optimisations** like **critical path CSS generation** and **single page application prerendering**. **Gulp.js**, **Ampersand.js**.

Led development of a holistic rewrite of the George Enescu festival website. Wrote scraping / data remodelling scripts to import old pages from the legacy PHP backend, added new capabilities like **full text search** and relationships between models. Built a **RESTful JSON API** for the newly developed native mobile apps. Deployed at scale with hundreds of thousands of hits per month. **Ruby on Rails**, **database and API design**, **load testing**, **New Relic monitoring**, **caching and database query performance optimisations**.

Projects

keepthingsopen.com (2026)

Co-wrote an open letter to NHS leadership asking them to keep their code open. It received over **2200 signatures** from notable industry leaders including Matt Hancock, Cory Doctorow, Dan Abramov, and Rich Harris.

tuicraft.vararu.org (2026)

A World of Warcraft chat client for the terminal, designed to be used by OpenClaw and other **AI agents** to play the popular MMO. Built with **Bun**, full test coverage, while heavily leveraging **AI code generation**. Open source, AGPLv3.

zp-voicemod (2026)

Reverse engineered the ZERO PARADES game and wrote a pipeline that trains a voice model (**hubert-bsqvae-small**) on a target voice and patches every sound file in the game with your desired voice. Open source, GPLv3.

untitled.ink (2025)

An interactive fiction platform featuring a Claude Code-style **agentic editor**, the **Ink** markup language, a hot-reloading preview, and video/image generation & editing on **Replicate** or **ComfyUI**. Not yet released.

The Last Question

A reimagined web presentation of Isaac Asimov's short story *The Last Question*.

Numeronym generator

A single-serving app that generates numeronyms (e.g., ally, i18n). Built using **Serverless** and **AWS Lambda**.

Space flight simulator

A 3D online space flight simulation game. **Meteor.js**, **Three.js**.

Dotfiles

Personal dotfiles and macOS configuration scripts, maintained over the years.